

Field Maintenance Summary —

FMS-2025-002

Document ID: FMS-2025-002

Reporting Period: 2025-05-01 to 2025-05-31

Site / Region: NW-PowerPool (WIND-024, WIND-014, WIND-003)

Field Supervisor: Diego Mendoza-Ruiz, Lead Field Supervisor, NW-PowerPool

Distribution: Regional Maintenance Manager (Karen Hsu); Operations Planning; Procurement Liaison; Safety Officer; on file at WIND-024 control building

Section 1 — Reporting Period Summary

May was the worst operating month I have had in this region in seven years. The Spring Vibration Cascade everyone is now referring to was not a cascade in the meteorological sense — it was the deferred PM bill coming due.

WO statistics, May 2025, NW-PowerPool:

Category	Planned	Completed	Deferred / Bumped	Emergency
Preventive WOs	27	14	13	n/a
Corrective WOs	18	11	7	n/a
Emergency WOs	0	n/a	n/a	3
Inspections	14	14	0	n/a

We completed less than half of our preventive WOs because the crew was diverted to emergency response on three separate weeks. Two of the deferred PMs are now corrective actions because their assets failed before we got to them.

Section 2 — Significant Events

5/2 — WIN-10058 soc_drift outage, NW-PowerPool-WIND-014.

GE turbine. State-of-charge drift on the auxiliary battery system tripped the unit at 14:22. Duration 227 hours. Lost generation 2,506.8 MWh. Full write-up in OIR-2025-006. Apex mobilized within four hours of CPS declining (they were two crews short, in May, in our region — this is what happens when you starve your scheduled vendor for four months). Apex invoice \$621,000. Their recommendation was

full battery-cabinet replacement. CPS, when I called them after the fact, said they would have refurbished it for under \$150K. Same scope on paper, very different bill.

5/19 — WIN-10011 vibration excess, NW-PowerPool-WIND-024.

This is the unit whose January PM was on the deferred list (WO-2025-00119).

Vibration excursion at 09:14. Took it offline at 10:01 controlled. Apex was on site by 18:00 the same day. Replaced two bearings. Invoice \$389,000.

5/26 — TRA-10561 vibration excess, NW-PowerPool-WIND-003.

Siemens transformer at the WIND-003 substation. Vibration excess outage at 03:11 — a 158-hour outage that cost us 2,615.1 MWh of lost generation. See OIR-2025-007.

Apex was on site by 11:00 with a Siemens Gamesa OEM subcontractor (the 1.8× rate uplift applies to that crew). They installed a vibration isolator and replaced two bushings. Invoice \$446,000. RAM-2025-001 documents the repair-vs-replace decision we deferred at the time.

Three Apex dispatches in one month. Total \$1,456,000. Average \$485,000. The average has gone up since January.

Section 3 — Vendor Activity

- **Apex:** Three dispatches, \$1,456,000. Their crew lead, Brad Holstein, is now on a first-name basis with my night dispatcher. That is not a relationship I wanted to develop.
 - **CPS:** Limited to scheduled work outside the outage response window. Maria Tonkin sent me a polite-but-firm note on 5/15 asking when we would resume normal scheduling. I do not have an answer for her. The freeze formally lifted in March but the backlog is still consuming us.
 - **Siemens Gamesa OEM service:** Two warranty inspections on WIN-10131 and WIN-10011, both with replacement recommendations totaling \$1.2M per unit. Both declined pending MOC-2025-022.
 - **Pinnacle:** One site visit for a previously scheduled scope on WIND-014. Their crew showed up with a service manual that referenced procedures from a 2019 predecessor-company manual. We had to stop work for half a day until the procedure conflict was resolved against our copy of the current Standard MS-2025-03.
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Section 4 — Open Issues / Backlog

- **Crew fatigue.** Three crew members have exceeded FMP-2's 60 hr/wk limit for two consecutive weeks. I filed a CAR (see CAR-2025-006 in process) on it because the policy says I have to. We have not had a safety event but the conditions are wrong.
 - **Repeat-failure pattern.** WIN-10058 has now failed twice this calendar year (1/29 inspection-flagged trend, 5/2 outage). WIN-10011 has failed twice (5/19, prior March event). WIN-10131 has had a January overheating event and Siemens Gamesa wants \$1.2M to replace it. These are not three separate stories; they are one story repeating at three turbines.
 - **Vibration baseline drift on WIN-10016.** Inspections in late May show baseline shifted approximately 12% above March reading. Standard MS-2025-03 threshold is 15%. We are inside the band but the trend is wrong.
 - **MOC-2025-022 still pending.** Two months in. Every Siemens Gamesa inspection produces a six-figure replacement recommendation we cannot evaluate against internal standard until this is resolved.
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Section 5 — Resource & Safety Notes

Crew of 11, plus two contractor technicians from CPS on rotation. Three FMP-2 exceedances logged (see CAR-2025-006). Two of those technicians had to take mandatory standdown weeks 5/27 and 6/3 — which dropped my available crew right when WIND-014 needed coverage.

Weather: a wet May. Eight days of restricted climbing. The 5/19 and 5/26 events both came after restricted-climbing days, which means inspection rounds we would otherwise have completed were pushed back.

One near-miss reported: 5/24, technician on WIN-10011 platform during pre-shutdown inspection lost footing on a wet grating. No injury. Logged.

Section 6 — Forward Look

June will be worse before it is better. We have nineteen corrective WOs queued — most of which originated as preventive WOs we did not complete in Q1 or May. We need either (a) a CPS surge — and Maria has told me she can give me eight extra technicians if we commit through Q3 — or (b) continued Apex dispatches at \$400K+ per event.

I am formally requesting through Karen that we authorize the CPS surge. It will cost us approximately \$1.1M for the quarter. We have already spent \$1.45M on Apex in May

alone. I would rather have CPS on site fixing things before they fail than Apex on call fixing them after.

I will be filing CAR-2025-006 on crew fatigue this week.

Submitted by:

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Lead Field Supervisor, NW-PowerPool
Cascadia Renewable Operations LLC

References:

- OIR-2025-006 — WIN-10058 soc_drift outage, 2025-05-02
- OIR-2025-007 — TRA-10561 vibration excess outage, 2025-05-26
- CAR-2025-006 — Crew fatigue, FMP-2 exceedances (in process)
- RAM-2025-001 — TRA-10561 repair-vs-replace decision (deferred)
- MOC-2025-022 — OEM Threshold Override (pending)
- FMS-2025-001 — January 2025 deferred PM list